25901167

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

DSC – MG4DSCCOM201- QUANTITATIVE TECHNIQUES FOR BUSINESS DECISIONS

Maximum Marks: 70 **Duration: 2 Hours**

Scheme for Valuation

Part A

Very Short Answer Questions. **Answer any 12 questions**.

Each question carries **1 mark**.

1. Linear Correlation
2. No maximum value
3. Rank Correlation
4. Standard Error of Estimate
5. Line of best fit also known as the trend line or regression line is a straight line that best represents the relationship between a set of data points.
6. Multiplicative Model
7. If two events A and B are independent, then the probability of both events occurring together (i.e., the intersection of A and B) is the product of their individual probabilities. $P(A \cap B) = P(A) \times P(B)$
8. 360
9. 75%
10. Classical Probability. Axiomatic Probability, Subjective Probability, Empirical Probability
11. Regression is called a predictive tool because it uses established mathematical relationships between variables derived from historical data to estimate unknown, continuous future values OR It shows how changes in one variable (independent) will impact another (dependent), allowing for quantitative forecasting in business
12. Irregular variation, or random variation, refers to unpredictable, short-term fluctuations in time series data that do not follow a set pattern, caused by erratic events like natural disasters, strikes etc.
13. 84
14. Bayes' Theorem

Part B

Short Answer Questions. **Answer any 4 questions**.

Each question carries **7 marks**.

15. Time series analysis is mainly used to explain, describe, and predict changes via the time of chosen variables. Many companies use time series forecasting, and analysis to develop

business strategies. These techniques help businesses in measuring, tracking, defining, and predicting business trends. The important points include:

Forecasting and Planning, *Trend and Seasonal Analysis*, *Anomaly Detection*: *Strategic Decision Making*, *Performance Evaluation* etc. *(Any other relevant points)*

16. Correlation measures the strength and direction of the linear relationship between two variables, indicating how they move together (e.g., height and weight). Its significance lies in identifying associations for prediction, decision-making, and modeling, with a range between -1 and +1.

No, correlation does not always signify a cause-and-effect relationship. While correlated variables move together, this does not mean one causes the other to change.. The coefficient of correlation ranges from -1 to +1:

+1 indicates a perfect positive linear relationship,

-1 indicates a perfect negative linear relationship,

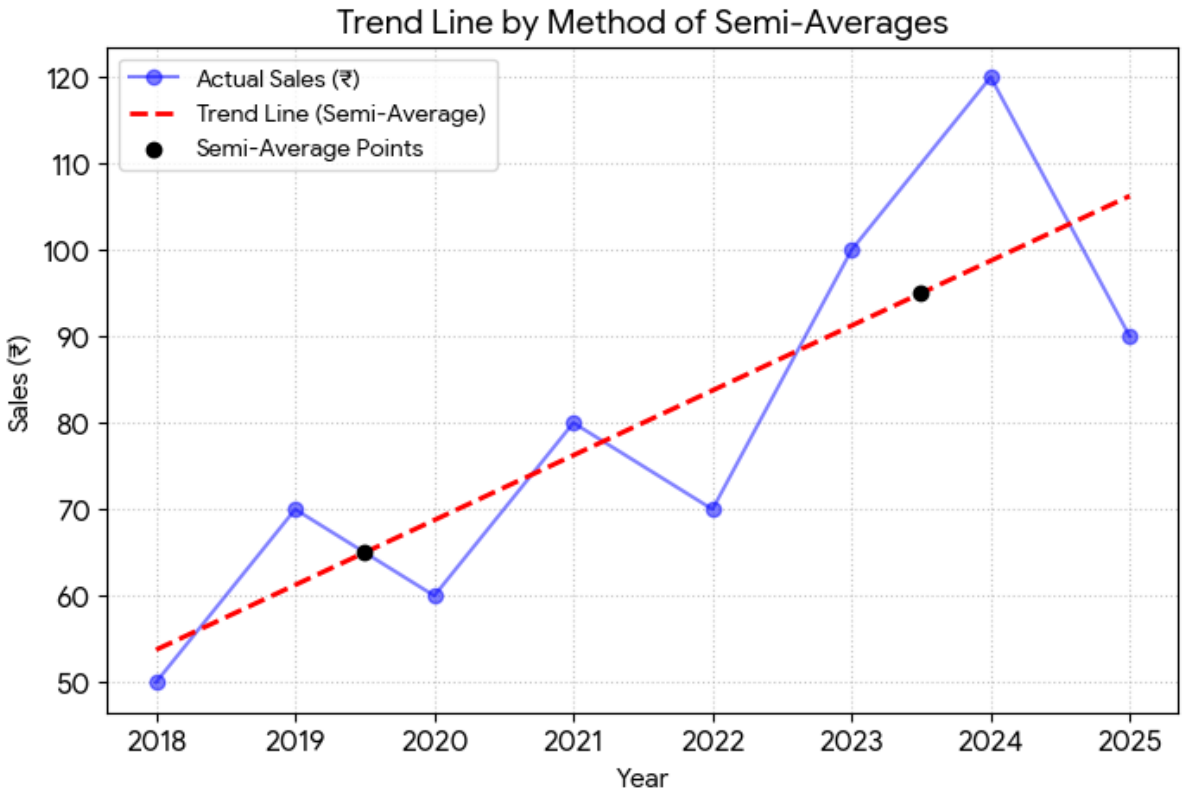
0 indicates no linear relationship.

17. To fit a trend line using the method of semi-averages, the data is divided into two equal halves, the average for each half is calculated, and a straight line is drawn through these two points.

Average for first four years 260/4 = 65

Average for last four years 380/4 = 95 (3 marks for dividing the series into two and calculating averages and 4 marks for the graph)

Middle year of first half 2019.5 and second half is 2023.5



18. The odds in favor of event A are given as 6 to 9

$$P(A) = \frac{6}{6+9} = \frac{6}{15} = \frac{2}{5} = 0.4$$

The odds against event B are given as 12 to 10

$$P(B) = \frac{10}{12+10} = \frac{10}{22} = \frac{5}{11} \approx 0.4545$$

- **Probability A fails:** $P(A') = 1 - P(A) = 1 - \frac{2}{5} = \frac{3}{5}$.
- **Probability B fails:** $P(B') = 1 - P(B) = 1 - \frac{5}{11} = \frac{6}{11}$.
- **Probability both fail:** $P(A' \cap B') = P(A') \times P(B') = \frac{3}{5} \times \frac{6}{11} = \frac{18}{55}$.

Finally, the probability the puzzle is solved is:

$$P(\text{solved}) = 1 - P(\text{both fail}) = 1 - \frac{18}{55} = \frac{55-18}{55} = \frac{37}{55}$$

19. (i) Exactly one stock from each sector = (3C1 X 4C1 X 2C1 X 1C1) / 10C4

= **24/210 = 4/35 or 0.114** (3 marks)

(ii) Auto stock and at least one banking stock =

If one auto stock is included (1C1= 1), number of selection for other stocks = 9C3= 84

If banking stock is excluded in the remaining (Pharma + IT only)= 6C3= 20

At least one banking stock = 84- 20 = 64

Therefore, number of inclusion of auto stocks and at least one banking stock

= **64/210 = 32/105 = 0.3047 (4 marks)**

20. Arithmetic Mean of X = 3 and Y = 3 (The student can apply any method including short cut formula)

r= 0.5 (Procedure and formula= 5 marks, Final answer = 2 marks)

Part C

Short Answer Questions. **Answer any 2 questions**.

Each question carries 15 **marks**.

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$$1 - \frac{6 \sum D^2}{N^3 - N}$$

R1	R2	R3	$D1^2$ (1 &2)	$D2^2$ (2 &3)	$D3^2$ (1 & 3)
5	8	4	9	16	1
2	5	3	9	4	1
3	1	2	4	1	1
8	2	7	36	25	1
10	9	8	1	1	4
1	4	1	9	9	0
4	7	5	9	4	1
6	3	6	9	9	0
9	10	10	1	0	1
7	6	9	1	9	4
Total			**88**	**78**	**14**

Note- The order can differ – (12 marks for calculations and 3 marks for interpretation)

R_{12} = 1 – (6* 88)/990 = .467

R_{23} = 1 – (6*78)/990 = .527

R_{13} = 1 – (6* 14)/990 = .915

The First and the Third Judges have the highest agreement as the rank correlation coefficient is positive and maximum in their judgement.

22 ***Note: The student can follow any method- This is only suggestive)***

X	X^2	Y	Y^2	XY
40	1600	32	1024	1280
34	1156	39	1521	1326
28	784	26	676	728
30	900	30	900	900
44	1936	38	1444	1672
38	1444	34	1156	1292
31	961	28	784	868
245	**8781**	**227**	**7505**	**8066**

Mean of X = 245/7 = 35 Mean of Y = 227/7 = 32.43

byx = ((7*8066)- (245*227)) /(7*8781) – (245^2) = .5873

Y – 32.42 = .5873X -20.55

Y = .5873X +11.8645

If X is 36 then Y is 33 (Y = (.5873*36) + 11.8645 = 33.02…)

(Formula, application and procedure = 10 marks, 5 marks for substitution and answer)

23 The estimated profit for MNO Ltd. in 2023 is ₹79.57 crores.

To fit a straight line trend using the Least Squares Method, we use the equation:

Yc = a +bx (y is profit, x is deviation from middle year 2019- *Note- deviations can be taken from any year*)

Substituting in the equations , a =55.71 and b =5.96. (Ignore approximation differences)

(a) Trend equation is **y= 55.71 + 5.96x**

Year	**Profit (Y)**	**x**	**(b) Trend Values**
2016	40	-3	37.82
2017	42	-2	43.79
2018	50	-1	49.75
2019	55	0	55.71
2020	60	1	61.68
2021	68	2	67.64
2022	75	3	73.61

(c) For the Year 2023 the value of x = 4

Y= 55.714+5.96*4 = **79.57 or approximately 80 crores**

(9 marks for the procedure and equation, 4 marks for trend values and 2 marks for forecast)